

ECD–QV Pipeline

Fock-buffer calibration (*cached*)
depth probe (*optional, --probe*)

$N_{\text{cav}}, k_{\text{start}}$

Gate Parameter Optimizer

ECDParameterFinder · L-BFGS (restart) / Adam
N Haar targets · ideal ECD model

$\{\beta_j, \theta_j\}$ per ECD layer

Pulse Builder

ECDPulseBuilder · solve $\alpha(t)$ ODE
virtual-Z correction · cavity phase

$\varepsilon(t), \omega(t)$ waveforms

Displaced Frame Simulation

DisplacedFrameSimulator · dynamiqs (default) / QuTiP
T1/T2 noise sweep · HOG & XEB metrics